GRASP-RAG: Path-Guided Reasoning (PGR) — Context Formats

One shared generation prompt; motif-specific structure lives in the **context**. Context builders: `_bridge_context`, `_single_entity_context`, `_multi_entity_context`; broad falls through to LightRAG hybrid.

(a) Bridge Query

`_bridge_context`

"What connects Rational Grazing to Polyface Farm?"

Path selected by edge-keyword relevance to query (not shortest-path only). Remaining chunk budget filled via graph-guided re-query using best path's keywords.

Description: Systematic approach to managing pasture through controlled grazing periods; maximises productivity and recovery. . .
[source passage. . .]

Keywords: livestock management, rotational grazing, animal health
[source passage. . .]

Description: Central to regenerative systems; Polyface Farm uses cows as primary animal in rotational schedule driven by rational grazing logic. . .
[source passage. . .]

Keywords: rotational livestock, farm operations, pasture management
[source passage. . .]

Description: Joel Salatin's regenerative farm in Virginia; multi-species rotational model with cows, pigs, and poultry. . .
[source passage. . .]

[Supporting Context] *(alternative paths + graph-guided re-query chunks)*
. . . additional evidence for the specific connection. . .

Effect: Bridge entity (**Cows**) is structurally unavoidable at Step 3; LLM must traverse the path to answer.

(b) Single-Entity Query

`_single_entity_context`

"What is Polyface Farm?"

Anchor = first node of first returned path. Direct connections drawn from all returned paths (up to 10). Secondary chunks filled by VDB re-query.

Type: organisation
Description: Polyface Farm is Joel Salatin's regenerative farm in Swoope, Virginia. Known for multi-species rotational grazing. . .

Connected to:
- **Joel Salatin**: ownership, farm management
- **Cows**: rotational livestock, primary grazer
- **Rational Grazing**: grazing method, pasture recovery
- **Chickens**: multi-species rotation, following cows
- **Holistic Management**: land stewardship, philosophy

Source Passages:
[source passage. . .] *[source passage. . .]* *[source passage. . .]*

Effect: LLM sees a rich entity profile plus all direct relations — optimised for definitional / factual single-entity questions.

(c) Multi-Entity Query

`_multi_entity_context`

"Compare Rational Grazing, Holistic Management, and Polyface Farm."

Unique nodes from all Steiner-tree paths collected as anchors. Per-entity profiles built first; connection map built from all path edges.

Description: Systematic pasture management through controlled grazing and recovery periods; maximises productivity. . .
[source passage. . .]

Description: Decision-making framework developed by Allan Savory; integrates social, economic, and ecological factors. . .
[source passage. . .]

Description: Joel Salatin's flagship regenerative farm; demonstrates multi-species rotation and integrated land management. . .
[source passage. . .]

Description: Primary grazing animal linking all three above entities. . .
[source passage. . .]

[Connections]
- Rational Grazing ↔ Cows: *livestock management, rotation*
- Holistic Management ↔ Cows: *herd impact, land regeneration*
- Cows ↔ Polyface Farm: *rotational livestock, operations*
- Rational Grazing ↔ Holistic Management: *land philosophy*

Effect: LLM sees per-entity profiles and an explicit connection map — optimised for comparison and multi-hop integrative questions.

(d) Broad Query

`LightRAG hybrid fallback`

"What are the key debates in regenerative agriculture?"

No custom PGR format. Code calls `rag.aquery(query, param=SearchParam(mode="hybrid"))`. Context is built by LightRAG: VDB chunk retrieval + keyword graph traversal, flat.

Passage 1: Regenerative agriculture encompasses practices like cover cropping, reduced tillage, and rotational grazing. . .
Passage 2: Critics argue that regenerative claims lack standardised certification and measurable carbon sequestration benchmarks. . .
Passage 3: Holistic Management's savanna hypothesis has been contested by ecologists who dispute the scalability of herd impact. . .

keywords: soil health, carbon sequestration, cover crops, rotational grazing, biodiversity, no-till farming, certification. . .

No step-by-step labels. No entity profiles. No connection map. Flat, unordered retrieval — same as LightRAG's own output.

Why: Broad queries have no specific entity anchors, so graph traversal motifs don't apply. GRASP-RAG delegates to LightRAG hybrid, which is optimised for diffuse thematic retrieval. Win rate vs LightRAG on broad queries is 69.6% — not structural.

Colour key: ■ = Entity step (blue) ■ = Entity profile (green) ■ = Supporting context (yellow) ■ = Connection step (orange) ■ = Multi-entity / connections (purple) ■ = Broad / flat (gray)

One shared prompt for all motifs (`_RESPONSE_PROMPT`). Structure is entirely in the context, not in the prompt instruction. Each motif's context builder imposes the topology; the LLM reads it as-is.